

Advanced Programming Techniques

fbeit

FACHBEREICH ELEKTROTECHNIK
UND INFORMATIONSTECHNIK

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h_da

HOCHSCHULE DARMSTADT
UNIVERSITY OF APPLIED SCIENCES

SICHERHEIT UND UMWELT



Brandschutz Information

- In case of fire -

Für IHRE Sicherheit

- For your own safety

Stand: 14.10.2022



1

Im Alarmfall wird das Gebäude komplett geräumt!

In case of an emergency the complete building will be evacuated

- S O F O R T -

- I M M E D I A T E L Y -



2

Fenster & Türen schließen – aber nicht verschließen!

Close windows & doors – but do not lock them



3

Hilflose, Hilfsbedürftige mitnehmen!

Guide helpless or disabled people to the assembly point



4

Der Fluchtwegkennzeichnung folgen!

Follow the signed escape routes





5

Aufzüge nicht benutzen!

Do not use elevators



6

Anweisungen der Brandschutzhelfer folgen!

Follow the instructions of the evacuation coordinators



7

Sammelplatz aufsuchen – auf Anweisungen warten!

Go to the assembly point – wait for instructions !



8

Nicht mit dem Auto vom Gelände fahren - Feuerwehrezufahrt wird blockiert!

Do not leave the campus with you car.

You block the access for the fire-service



9

Rückkehr in das Gebäude erst nach Freigabe durch die Feuerwehr-Einsatzleitung!

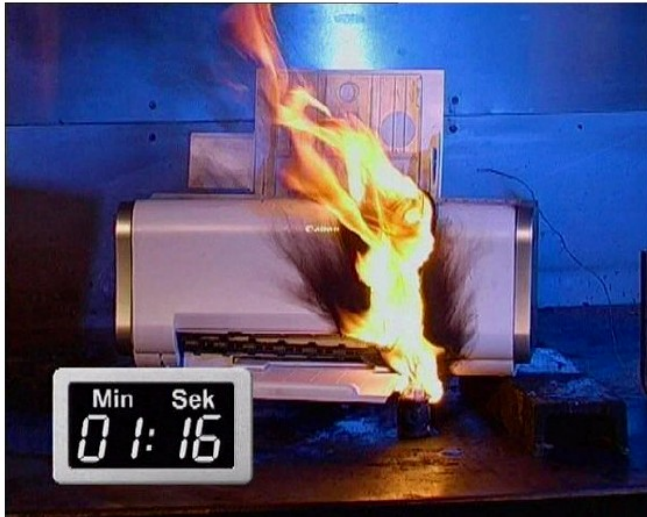
No return into building before fire chief's decision !



10

Es gibt keine Spaß - Alarmierung!

There will be never a fake fire alert



Sicherheitseinrichtung – Safety equipment

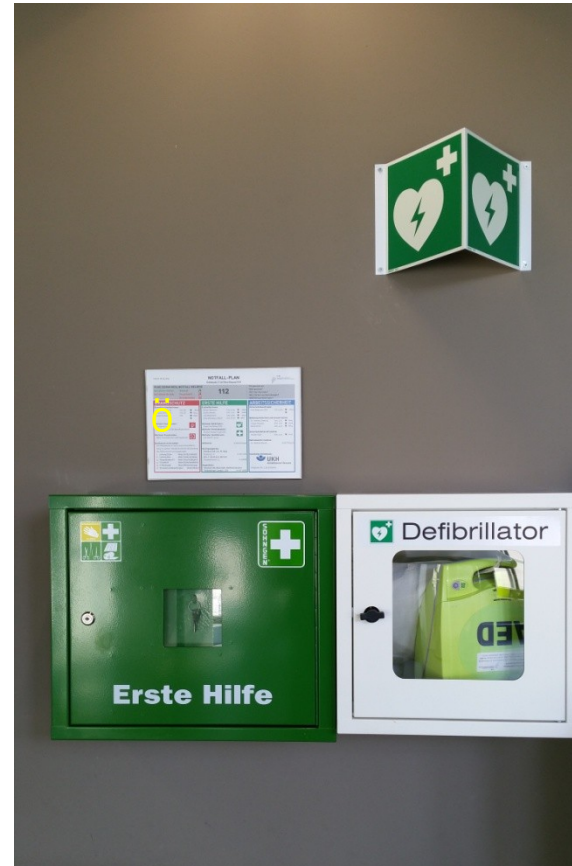


Erste Hilfe Koffer &
Notfallplan in jeder
Etage



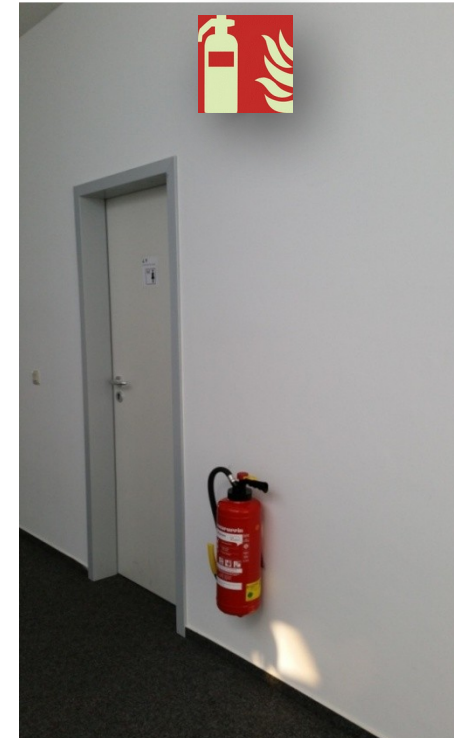
First aid kit &
emergency plan in
every floor

Erste Hilfe Koffer, Notfall-
plan & Defibrillator (AED)
an zentraler Stelle



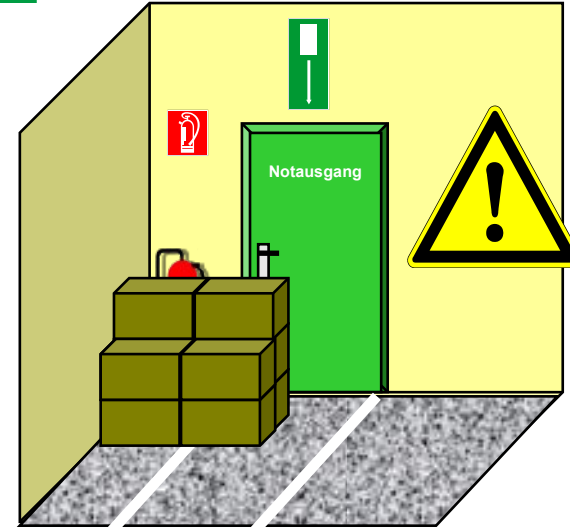
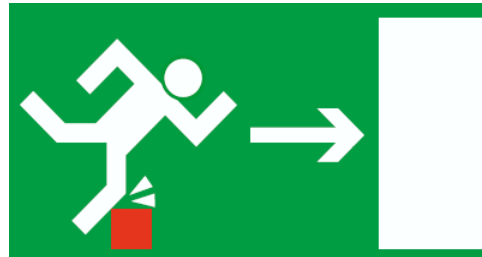
First aid kit, emergency plan
& defibrillator (AED)
at central point

Feuerlöscher
überall



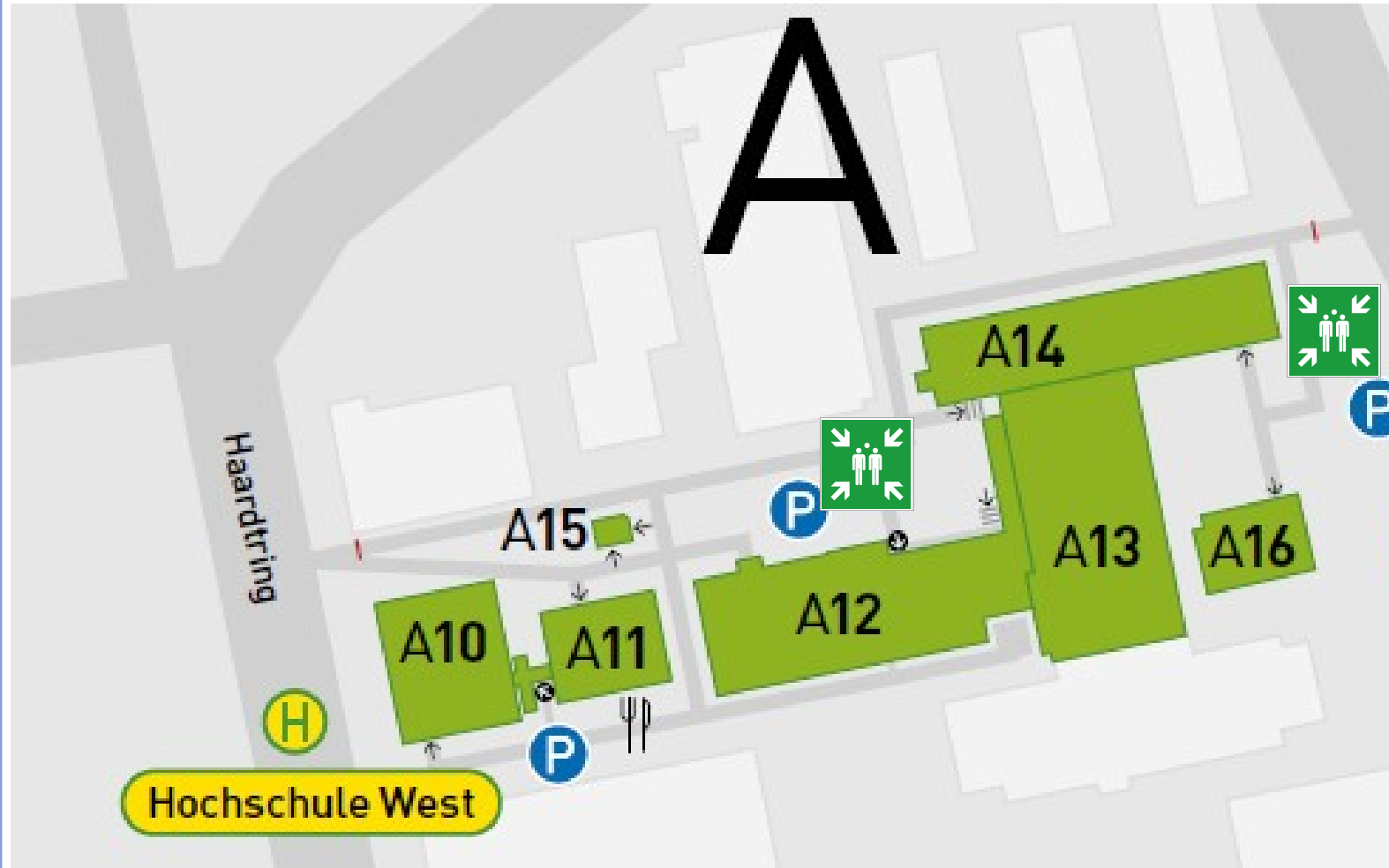
Fire extinguisher
everywhere

Für IHRE Sicherheit - For your own safety



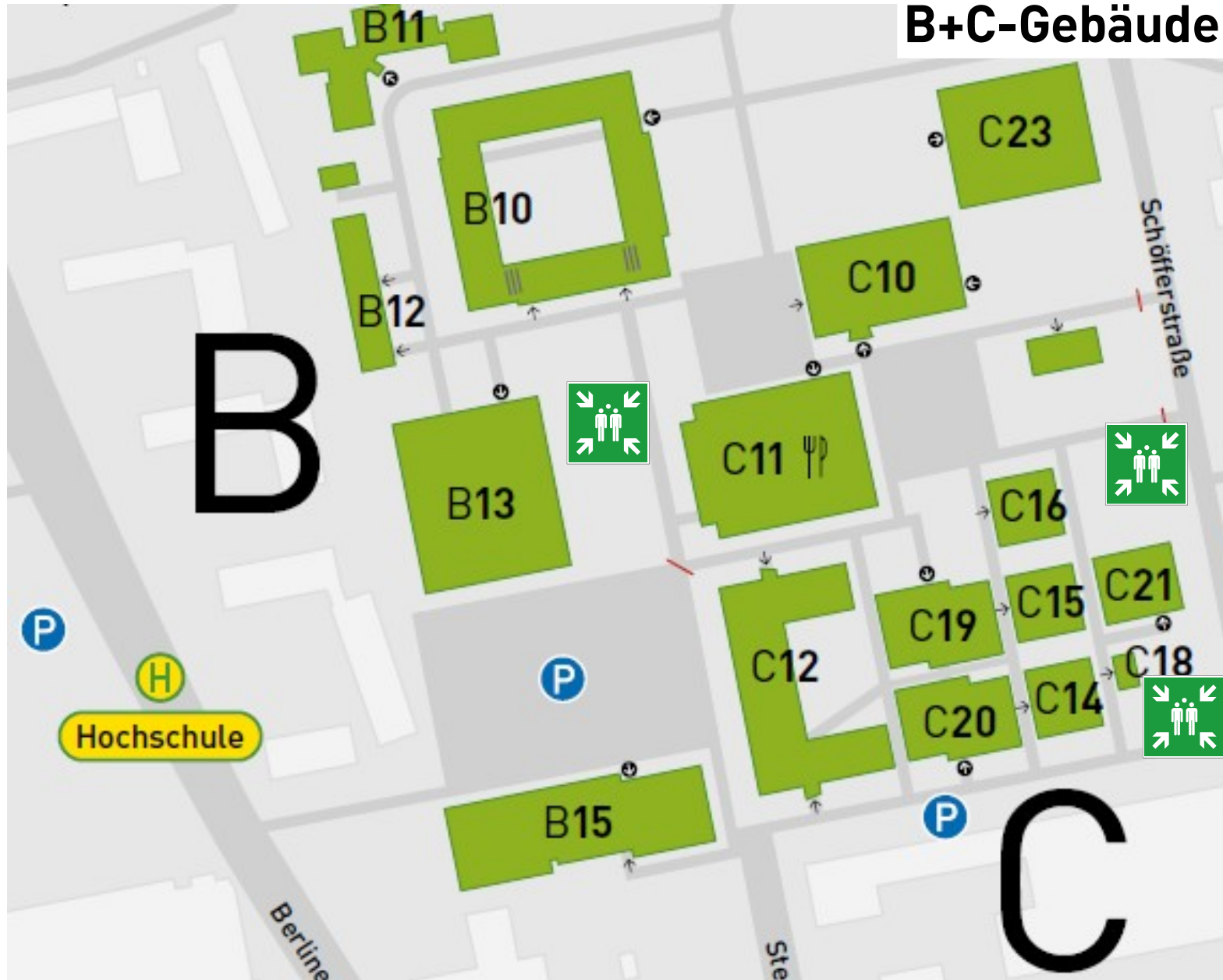
Sammelplätze – Assembly Points Campus Darmstadt

A-Gebäude



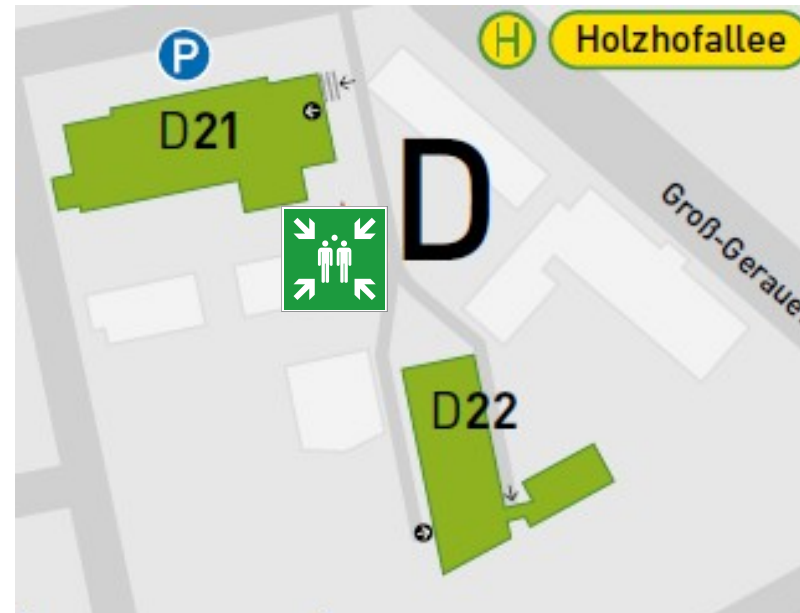
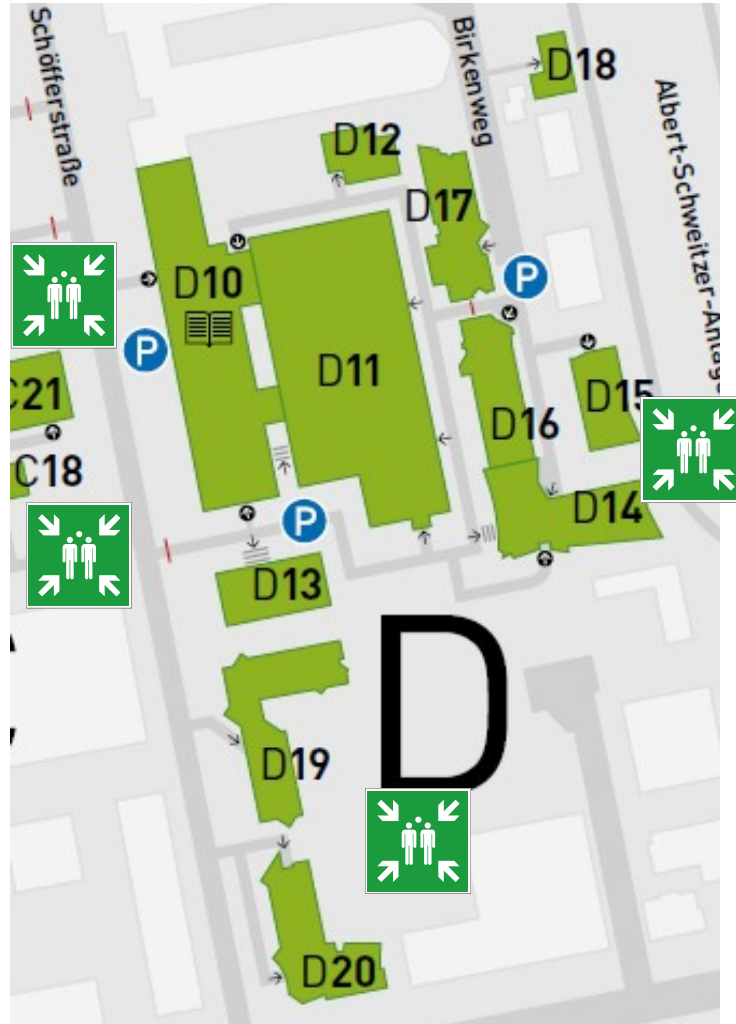
Sammelplätze – Assembly Points Campus Darmstadt

B+C-Gebäude



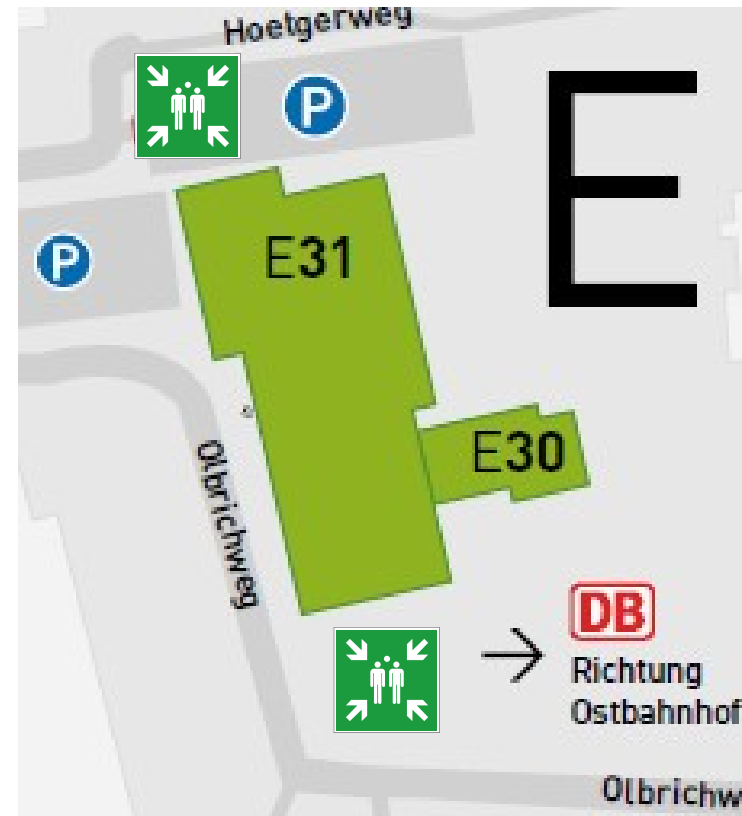
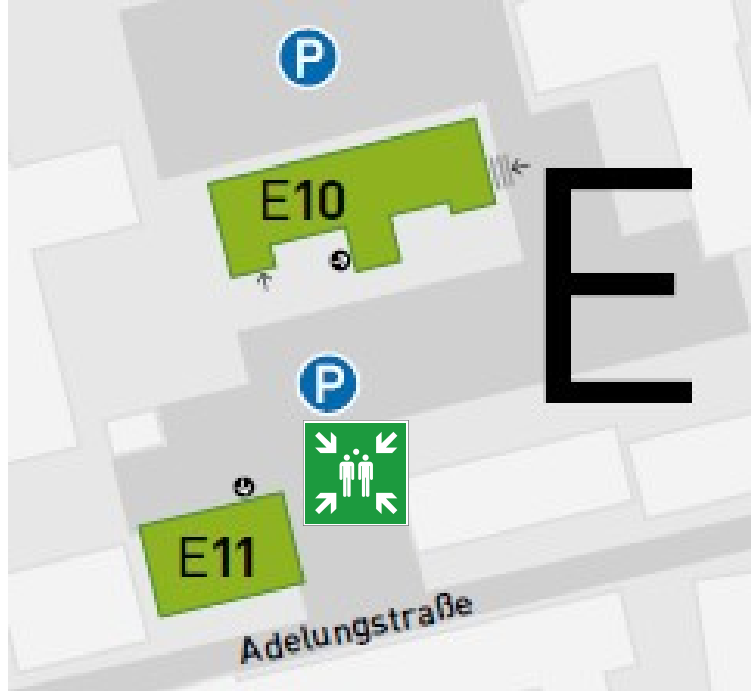
Sammelplätze – Assembly Points Campus Darmstadt

D-Gebäude



Sammelplätze – Assembly Points Campus Darmstadt

E-Gebäude



Sammelplätze – Assembly Points Campus Dieburg



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Organisation

Don't distract you fellow students



Goal of this Module

- **Learn about Object Oriented Programming (OOP)**
 - Class design and class relationships
 - Design patterns and algorithms
- **Practice OOP using an Object Oriented System-Level Programming Language (C++)**
 - Design & Implementation
 - Memory management
 - Operator overloading
 - ...
- **Systematic Software Development**

Prerequisites for this Module

- **Knowledge of the C-Programming language**
 - Data types, variables, statements, programs
 - Operations
 - Control statements
 - Enum, struct
 - Pointers
 - Functions and parameters
 - Standard library (basics)
 - Preprocessor basics (`#include`, `#define`)
- **... and how to use it!**

Lecture and Lab

- **Module consist of a lecture and a lab**
- **Lecture provides a “guided tour”**
 - Consider it a starting point
 - Most topics need practice for real understanding
- **Lab provides practical experience**
 - Programming tasks
 - Open labs every week (make use of them!)
 - Three review labs
- **Exam**
 - Practically solve a programming problem (“Computer exam”)

Some of you will have a steep learning curve. Use the Open Lab!

Material from the Lecture

- **Material is made available in h_da's Moodle**
 - <https://lernen.h-da.de>
 - Course name "MA04 Advanced Programming Techniques – Lecture"
 - Course Short name "APT-L" + year
 - Freely accessible, no enrollment key
 - Link to Eclipse download
 - "Customized" version that includes everything required for Windows 10/11 (64-bit)
 - Sorry, absolutely no MacOS support
 - Using Linux is okay (I do it), but you're on your own
 - Includes project template with setting for course

Lab Organization

- **Exercises are made available in Moodle course**
 - Course name “MA04 Advanced Programming Techniques – Lab”
 - Short course name “APT-P” + year
- **Enrollment by teacher for regular participants**
- **Exercises**
 - Will be made available successively at specific dates (see Moodle course)
 - Approx. 3 weeks for completion (make use of open labs!)
 - Submission until specified date
 - Pre-check (compilability, **check for plagiarism**)
 - Mandatory “review lab” for getting mark
- **Lab has to be passed as prerequisite for exam**

Additional Material

- **Online**

- <https://en.cppreference.com/w/>
- <http://www.cplusplus.com/>
- <https://www.tutorialspoint.com/cplusplus/>
- Lots of additional resources, Google is your friend

- **Printed**

- Bjarne Stroustrup, The C++ Programming Language, 4th edition
- Other books (C++14)